

Markov Chain Monte Carlo

Swastika Dey

INSTRUCTOR: Manjunath Krishnapur, IISc Bangalore

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Introduction

In many statistical contexts, it becomes essential to sample from a given distribution. For example, we might be interested in computing an integral. The classical Monte Carlo approach tells us to interpret this integral as an expectation and generate an *iid* sample from the corresponding distribution, whose empirical mean is a good approximation for the expectation (due to the strong law of large numbers). But in practice, sampling from the distribution may be difficult! In some cases, the density might be known up to some constant i.e., a function f that satisfies $f \geq 0$ and $\int_{-\infty}^{\infty} f(x) dx < \infty$ is known, but computing $\int_{-\infty}^{\infty} f(x) dx$ is impossible. This particularly arises in Bayesian statistics. The existence of simulation algorithms in such cases allows one to sample from the posterior distribution of some complicated model. In addition to the statistical applications, Markov chain Monte Carlo algorithms are of importance in statistical physics wherein one has to sample from certain spin models.

The way an MCMC algorithm goes is the following. Given a (possibly unnormalised) target distribution π , a Markov chain is constructed whose stationary distribution is the normalized version of π . The convergence of this Markov chain, or its running average to the stationary distribution, is given by certain ergodic theorems. This treatment requires the theory of Markov chains on a **general state space**, such as some euclidean space $\mathbb{R}^d$. However, for simplicity, in these notes, we restrict ourselves to Markov chains on a **countable state space** and hope that the MCMC algorithms work thus in the general case as well.

1 The Metropolis-Hastings algorithm

Let Q be the (known) transition matrix of an irreducible Markov chain. Choose X_0 . For $n \geq 1$, define X_n recursively as follows:

1. Generate Y_{n+1} from $Q(X_n, \cdot)$. That is, given $X_n = x$, $\mathbb{P}(Y_{n+1} = y) = Q(x, y)$.
2. Flip a coin with head independently with probability $\alpha(X_n, Y_{n+1})$ (to be defined later). If the result is head, then set $X_{n+1} = Y_{n+1}$, otherwise set $X_{n+1} = X_n$.

Clearly, X_n is a Markov chain with transition probabilities

$$P(x, y) = \begin{cases} \alpha(x, y)Q(x, y) & \text{if } y \neq x \\ 1 - \sum_{z \neq x} \alpha(x, z)Q(x, z) & \text{if } y = x. \end{cases}$$

Setting

$$\alpha(x, y) = \min\left\{1, \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)}\right\},$$

we see that the Markov chain X_n is time-reversible with stationary distribution π (normalised), i.e.

$$\pi(x)P(x, y) = \pi(y)P(y, x).$$

2 Glauber dynamics

The Glauber dynamics is an algorithm for simulating from a high-dimensional probability measure, such as that on a graph. Suppose V is a set (think of it as the set of vertices of a graph) and S is a finite set (interpreted as colours). Our objective probability distribution is a possibly unnormalised measure π of S^V . The elements of S^V are called **configurations** (think of them as graph colourings). Now we describe the Glauber dynamics on this state space. In statistical contexts, this algorithm is known as the **Gibbs sampler**.

We want to create a Markov chain on S^V whose stationary distribution is π (normalised). Suppose that the present state is $x \in S^V$. A vertex v is chosen at random from V . The new state is then selected according to the distribution π conditioned on the set of configurations that keep the state of all other vertices unchanged. More precisely, let

$$\Omega := \{y \in S^V : y(\omega) = x(\omega) \forall \omega \neq v\}.$$

Note that Ω depends both on the randomly chosen vertex v and the present state x . Then, the new state is chosen according to the distribution

$$\pi(y|\Omega) := \begin{cases} \frac{\pi(y)}{\pi(\Omega)} & y \in \Omega \\ 0 & y \notin \Omega. \end{cases}$$

It is not difficult to check that the generated sequence is a Markov chain and has the normalised version of π as the stationary distribution.

We conclude by presenting an application of this algorithm in the case of a well-known interacting particle system in statistical physics, the **Ising model**.

The Ising model

The Ising model is a popular **spin system**, used to describe the magnetic behaviour of a substance. It specifies a probability measure on $\{-1, +1\}^V$, where V is the set of vertices of a graph $G = (V, E)$. The vertices represent dipoles, -1 and $+1$ being the two possible directions of each dipole.

For any spin configuration $\sigma \in \{-1, +1\}^V$, the **energy** (or **Hamiltonian**) of the configuration is defined by

$$H(\sigma) := - \sum_{\{u,v\} \in E} \sigma(u)\sigma(v).$$

Note that if more spins are aligned in the same directions, the energy of the system is less. The **Gibbs measure** corresponding to energy H , is the following.

$$\mu(\sigma) := \frac{1}{Z(\beta)} e^{-\beta H(\sigma)},$$

where β is a parameter known as the **inverse temperature** and $Z(\beta)$ is the normalising constant, also known as the **partition function**. It is infeasible to compute the partition function $Z(\beta)$ for graphs with a large number of edges. We therefore resort to the MCMC algorithm described previously.

The Glauber dynamics for the target distribution μ works as follows. Let the current state of the chain be σ . A vertex v is chosen at random from V . Then the new state is chosen according to the measure μ conditional on the spins at all other vertices remaining the same. More precisely, the conditional probability of the spin at v being $+1$ is

$$\frac{e^{\beta S(\sigma,v)}}{e^{\beta S(\sigma,v)} + e^{-\beta S(\sigma,v)}} = \frac{1 + \tanh(\beta S(\sigma,v))}{2},$$

where $S(\sigma, v) = \sum_{u:(u,v) \in E} \sigma(u)$. This chain has the Gibbs measure as its stationary distribution.

3 Coupling From The Past:

We already saw how Markov chains can be useful algorithmically. If we have a probability distribution we would like to generate random samples from, we design an *ergodic* Markov chain whose unique stationary distribution is the desired distribution. We then run the chain (i.e., start at an arbitrary state and evolve according to the transition matrix), until the process is at (or close to) the stationary distribution.

We know that the chain eventually reaches the stationary distribution. In order for this Markov chain approach to be efficient, we need to know how fast we converge to stationarity. Now we will look at a method which notices when we reach the stationary distribution. The resulting samples are guaranteed to be from the stationary distribution and in many cases the algorithm is extremely efficient. This method is known as **Coupling from the Past**.

3.1 Intuition:

Let M be an ergodic Markov chain defined on a space Ω with transition matrix P . Thus, for $x, y \in \Omega$ and integer $t > 0$, $P^t(x, y)$ is the probability of going from x to y in t steps. Let π be the stationary distribution of M . Consider $\Omega = \{A, B, C\}$ and $P(A, A) = P(A, B) = P(C, A) = P(C, C) = \frac{1}{2}$ and $P(B, C) = 1$. Define a “global” coupling for the three chains, i.e; a joint evolution for Markov chains starting from all 3 states, such that each chain viewed in isolation is a faithful copy of the chain of interest. Now, as soon as all three chains reach the same state (i.e., they’ve all coupled), output the resulting state. **Is the output from the stationary distribution?**

We know that once we’ve coupled from all pairs of initial states, we’ve reached stationarity. Thus, we might expect that the output is from stationarity. However, this is **false**. In our example the output will never be state B, because B is never the first state where all the chains couple as B has a unique predecessor A and hence the chains would have coupled a step earlier at state A. But B has some positive probability in the stationary distribution, so the output is clearly not from the stationary distribution. However, it turns out that if we run this experiment “backwards”, it works. It’s a very clever idea. Let’s formalize the notion of a global coupling before detailing the general algorithm and analyze it.

3.2 Formal Description of The Method:

Let $f : \Omega \times [0, 1] \rightarrow \Omega$ be a *global coupling*, i.e., for all $x \in \Omega$ and for r chosen uniformly at random from $[0, 1]$, probability that $f(x, r) = y$ holds equals $P(x, y)$. Thus, the random seed r defines the evolution from all states, i.e; from $X_t \in \Omega$, the chain evolves by choosing r uniformly at random from the interval $[0, 1]$ and then moving to state $f(X_t, r)$. Let $\{r_t : t \in \mathbb{Z}\}$ be independent random variables, each chosen uniformly at random from $[0, 1]$. Define $f_t : \Omega \rightarrow \Omega$ by $f_t(x) = f(x, r_t)$. For a chain (X_t) , the function f_t determines the evolution at time t , i.e; $X_{t+1} = f_t(X_t)$. Let $t_1 < t_2$ be integers.

Define $F_{t_1}^{t_2}(x) = (f_{t_2-1} \circ f_{t_2-2} \circ \cdots \circ f_{t_1})(x) = f_{t_2-1}(f_{t_2-2}(\cdots f_{t_1}(x) \cdots))$. Thus, $F_{t_1}^{t_2}$ defines the evolution of the chain from time t_1 to time t_2 . Hence $\forall x, y \in \Omega$, probability that $F_{t_1}^{t_2}(x) = y$ holds equals $P^{t_2-t_1}(x, y)$. The earlier algorithm, which simulated the chains forward in time, can now be formalized as follows: We will run $|\Omega|$ chains X^σ , one for each $\sigma \in \Omega$, simultaneously. Each chain X^σ starts from its corresponding initial state, i.e., $X_0^\sigma = \sigma$. All these chains will evolve simultaneously using the coupling function f and we will get the final state as output once they have *coalesced*. More formally, let T be the first time such that $|F_0^T(\Omega)| = 1$. Observe that F_0^T is a constant function, i.e., $|F_0^T(\Omega)| = 1$ which means that all chains have reached the same state. Then the output value is the unique element of the set $F_0^T(\Omega)$. We now ask the following question: **Will the output value be drawn from the stationary distribution?**

It turns out from our example on the 3-state chain that the output value from the previous algorithm does **not** necessarily come from the stationary distribution. This motivates a rather strange question: what would happen if we **run the chain from the past**? More formally, let M be the first time such that $|F_{-M}^0(\Omega)| = 1$. Then if we output the unique element of $F_{-M}^0(\Omega)$, **what is the distribution of this output?**

The following theorem gives answer to this.

Theorem 3.1. F_{-M}^0 has the same distribution as π , the stationary distribution.

Proof. For any fixed $t > 0$, note that for all $x, y \in \Omega$,

$$\Pr(F_0^t(x) = y) = \Pr(F_{-t}^0(x) = y).$$

The probability space for the left-hand side is over the random choices $r_1, r_2, \dots, r_t$, whereas for the right-hand side it is over the random choices $r_{-t+1}, \dots, r_{-1}, r_0$. Thus,

regardless of the order in which we construct these t random seeds, the distributions of F_0^t and F_{-t}^0 are identical. Therefore, if we run the simulation infinitely far into the past, we reach stationarity, i.e; for all $x, y \in \Omega$,

$$\lim_{t \rightarrow \infty} \Pr \left(F_{-t}^0(x) = y \right) = \lim_{t \rightarrow \infty} \Pr \left(F_0^t(x) = y \right) = \pi(y).$$

For $t = t_1 + t_2$, observe that we have, $F_{-t}^0 = F_{-t_2}^{-t_1-1} \circ F_{-t_1}^0$. Therefore, if F_{-M}^0 is a constant function, then for all $t > M$ and all $x \in \Omega$,

$$F_{-t}^0(x) = \left(F_{-t}^{-M-1} \circ F_{-M}^0 \right)(x) = F_{-M}^0(x).$$

Hence, outputting $F_{-M}^0(x)$ is equivalent to outputting $F_{-\infty}^0(x) = \pi(x)$. This proves the theorem. $\square$

Remark: The above proof captures the essential difference between running the chain **backwards** versus **forwards** in time. Recall that T is the first time t such that F_0^t is a constant function. For $t > T$, we know that F_0^t remains a constant function, but it is not necessarily true that $F_0^t(x) = F_0^T(x)$, i.e; after time T , all couplings have coalesced, but the specific state they coalesce to may still change as t increases. In contrast, as shown above, for $t > M$ we have $F_{-t}^0(x) = F_{-M}^0(x)$, so the chains continue to coalesce upon the **same** state.

The above theorem translates into an algorithm for **perfect sampling** from the stationary distribution of the Markov chain M . However, there are several issues we must address before obtaining an efficient sampling algorithm:

- (a) **Implementation complexity:** Is it possible to implement the algorithm efficiently? More precisely, can we decide whether $|F_{-t}^0(\Omega)| = 1$ in polynomial time, even though the sample space of the Markov chain may be exponentially large?
- (b) **Choice of coupling:** How do we choose the coupling function f ? We would like to make $\mathbb{E}[M]$ small. Note that for a poor choice of f , it may even happen that $\Pr(M = \infty) > 0$.

It turns out that for **monotone systems**, we can address both of the above challenges effectively. We now consider the following example, which demonstrates that in the case of the **Ising model** (which is a monotone spin system), we can use the natural partial ordering of the model to decide coalescence efficiently by computing F_{-t}^0 for just two elements $\top, \perp \in \Omega$. Moreover, the expected running time $\mathbb{E}[M]$ will be

$O(T_{\text{mix}}(1/4) \ln |V|)$, where $T_{\text{mix}}(1/4)$ denotes the **mixing time** of the chain, i.e; the smallest time (starting from the worst initial state) such that the total variation distance to the stationary distribution is at most $1/4$.

Ising Model Example:

We already described the Ising Model structure in Glauber Dynamics. The transitions $X_t \rightarrow X_{t+1}$ are defined as follows:

1. From $X_t \in \Omega$, choose $v \in V$, $s \in \{+1, -1\}$, $r \in [0, 1]$, uniformly at random.
2. Define a candidate state X' by

$$X'(v) = s, \quad X'(w) = X_t(w) \text{ for all } w \neq v.$$

3. Set

$$X_{t+1} = \begin{cases} X', & \text{if } r \leq \min \left\{ 1, \frac{w(X')}{w(X_t)} \right\} = \min \left\{ 1, e^{-\beta H(X')} / e^{-\beta H(X_t)} \right\}, \\ X_t, & \text{otherwise.} \end{cases}$$

There is a natural partial order on Ω defined as follows: for $\sigma_1, \sigma_2 \in \Omega$, we have $\sigma_1 \preceq \sigma_2 \Leftrightarrow \sigma_1(v) \leq \sigma_2(v) \quad \forall v \in V$. Let $\perp$ and $\top$ denote the minimum and maximum elements under $\preceq$, respectively. Let f be the coupling in which v , s , and r are the same for all chains. Then $X_t \preceq Y_t \implies X_{t+1} \preceq Y_{t+1}$, i.e., the coupling preserves the ordering.

To see this, note that

$$\frac{e^{-\beta S(X', v)}}{e^{-\beta S(X_t, v)}} = e^{\beta(s - X_t(v)) \sum_{\{u, v\} \in E} X_t(u)}.$$

Thus, for $s = -1$, if the chain Y_t makes the move, then X_t also does; hence $X_{t+1} \preceq Y_{t+1}$. A similar argument holds for $s = +1$. Since f preserves monotonicity, $|F_{t_1}^{t_2}(\Omega)| = 1 \iff F_{t_1}^{t_2}(\perp) = F_{t_1}^{t_2}(\top)$. The **coupling time** of f is the smallest T such that $F_0^T(\perp) = F_0^T(\top)$ and the **coupling-from-the-past time** of f is the smallest M such that $F_{-M}^0(\perp) = F_{-M}^0(\top)$.

We have

$$\Pr(T > t) = \Pr(F_0^t(\perp) \neq F_0^t(\top)) = \Pr(F_{-t}^0(\perp) \neq F_{-t}^0(\top)) = \Pr(M > t).$$

Hence, M and T have the same distribution.

Let $\bar{d}(t) = d_{\text{TV}}(P^t(\perp, \cdot), P^t(\top, \cdot))$. Then $\bar{d}(t) \leq 2 \max_{z \in \Omega} d_{\text{TV}}(P^t(z, \cdot), \pi)$.

Thus, if we can show that $T_{\text{mix}}(\varepsilon) \leq t$, then it follows that $\bar{d}(t) \leq 2\varepsilon$.

Theorem 3.2. $\Pr(M > t) = \Pr(T > t) \leq \bar{d}(t)|V|$.

Proof. For $z \in \Omega$, let $h(z)$ denote the length of the longest decreasing chain under the partial order $\preceq$ that starts with z .

$$h(z) = \max\{k : \exists z = z_0 \succ z_1 \succ \cdots \succ z_k\}.$$

If $X_t \succ Y_t$, then $h(X_t) \geq h(Y_t) + 1$. Now, let $X_0 = \perp$, $Y_0 = \top$. Then

$$\begin{aligned} \Pr(T > t) &= \Pr(X_t \neq Y_t) \\ &\leq \mathbb{E}(h(Y_t) - h(X_t)) \\ &= \mathbb{E}(h(Y_t)) - \mathbb{E}(h(X_t)) \\ &\leq |V| d_{\text{TV}}(P^t(\perp, \cdot), P^t(\top, \cdot)) \\ &= |V| \bar{d}(t), \end{aligned}$$

as required. □

Theorem 3.3. $\mathbb{E}[M] = \mathbb{E}[T] = 2 T_{\text{mix}}(1/4) \ln(4|V|)$.

Proof. Let $t = T_{\text{mix}}(1/4) \ln(4|V|)$. Note that $T_{\text{mix}}(1/(4|V|)) \leq t$. Hence, $d(t) \leq \frac{1}{2|V|}$.

Therefore,

$$\Pr(T > t) \leq \frac{1}{2} \quad \text{and hence} \quad \Pr(T > kt) \leq \frac{1}{2^k}.$$

Thus,

$$\mathbb{E}[T] = \sum_{\ell \geq 0} \Pr(T \geq \ell) \leq \sum_{k \geq 0} t \Pr(T \geq kt) \leq \sum_{k \geq 0} t \frac{1}{2^k} = 2t.$$

□

Thus our proof for Ising Model completes.

4 Conclusion:

One may read [1] for more on countable MCMC. For general MCMC go through [3] and the paper [2] has more detailed discussions on coupling from the past.

References

- [1] David A Levin **and** Yuval Peres. *Markov chains and mixing times*. **volume** 107. American Mathematical Soc., 2017.
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- [3] Gareth O Roberts **and** Jeffrey S Rosenthal. "General state space Markov chains and MCMC algorithms?" *in* *Probability Surveys*: 1 (2004), **pages** 20–71.